Report 3: Natural Language Processing

Note: Key figures from the notebook are included here. Reference the notebook for further descriptive statistics.

Section 1: Exploratory Data Analysis, Data Preparation, and Sentiment Landscape

The US Airlines "Tweets" dataset comprises 14,640 customer tweets spanning across six major US airlines, and 14,427 unique tweets remained for analyses after removing duplicate entries and cleaning URLs, special chars, whitespace, and case normalization. The sentiment distribution skews negative, as 62.9% of tweets are labeled negative compared to 21.2% neutral and 15.9% positive. From initial findings, we can surmise that customers are more likely to post when experiencing failures in service than when service meets or exceeds expectations.

Sentiment distribution varies across airlines, with perception of some outperforming others due to factors like operational reliability (Figure 1A). US Airways (77.8% negative) and American (71.9%) have the highest share of negative tweets, with United (69.2%) following next. By contrast, Virgin's sentiment profile is more balanced (36% negative, 30% positive), and Delta and Southwest have lower shares of negative sentiment (43.2% and 49.2%, respectively).

From the data, it can be inferred that Twitter acts as a mechanism for bringing to light issues that remain unresolved. Airlines with higher proportions of negative comments may be dealing with more breakdowns (i.e., delays, cancellations, poor customer service) at a higher frequency, prompting discussion on public platforms like Twitter. In general, tweets tend to be short but communicate a clear message. The average tweet is 16.6 words (87.4 characters), but emotional signals are clearly communicated (Figure 1B, 1C).

For product and operations teams, social media represents a hub of expressing breakdowns and negative sentiment more so than neutral samples. Teams should constantly monitor platforms such as Twitter as early indicators of distress to be handled. Doing so can present service gaps that can further escalate in brand damage.

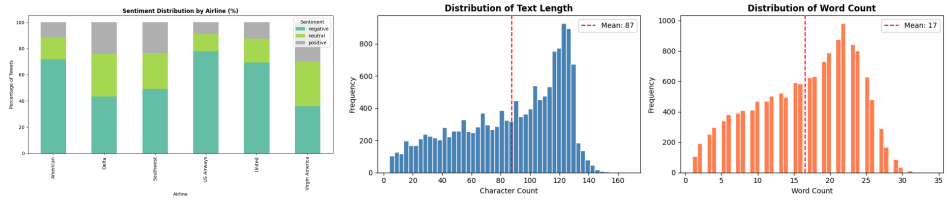

Figure 1. A: Sentiment by airline (left), B: Text length distribution (center), C: Word count distribution (right).

Section 2: Reason and Theme Analysis

Human encoding labeled 9,178 tweets (62.7%) as negative, although 2,602 tweets (28.4%) were considered ambiguous (represented as "Can't Tell" or low confidence). These entries were enriched using ChatGPT 5.2. LLM to improve or fill in missingness from original complaints. The primary drivers of dissatisfaction include customer service issues (31.7%), late flights (18.1%), cancelled flights (9.2%), lost luggage (7.9%) and bad flight experiences (6.3%). Nearly one-third of all negative tweets are customer service complaints, meaning that gripes with the airlines not only exist operationally, but for how breakdowns are handled. Subsequent complaints are tied to scheduling and delays.

Among ambiguous tweets, the LLM confidently relabeled 461 tweets (5% of human-identified negative tweets). Agreement between the original human encoding and the LLM's suggestions remained low at 24.3%, but is expected given that ambiguous cases are typically low confidence. The average LLM confidence score for these cases was 0.37, meaning that many tweets had limited signals or overlapping

complaints that were difficult to parse. Contextually, it could mean that complaints are driven by a multitude of factors (e.g., “my flight was late and no one helped me”) and strict categorization into defined buckets could be limiting. Critically, enrichment reduced the “Can’t Tell” category by 216 cases and a new “Other” category with 28 tweets was introduced.

The LLM-generated labels appeared generally reasonable for clearly stated issues (e.g., “flight delayed 3 hours” → Late Flight) where there were strict and concrete observable events. However, it struggled more with vague dissatisfaction or multi-issue tweets, as indicated by the low agreement score mentioned above. For product teams, it is important for them to understand that issues are often not siloed, but rather a compounding of multiple past negative experiences.

Reason distribution varies by airline. For American and US airways, customer service issues comprise 36-39%, late flight share is 28.6% for Delta, United is characterized by higher share of lost luggage (10.1%) and Virgin is the most balanced (Figure 2A). Different carriers possess different pain points, with some struggling with reliability while others can improve communication. Temporally, negative tweet volume peaks between 8AM to 3PM, which matches typical travel hours (Figure 2B). Customer service issues are consistent across all hours, while delay-related complaints increase during high-traffic windows (which could indicate that people tweet frustration in real time. While operational reliability is important, the data suggests airline teams should respond to issues with customer service rather than just punctuality.

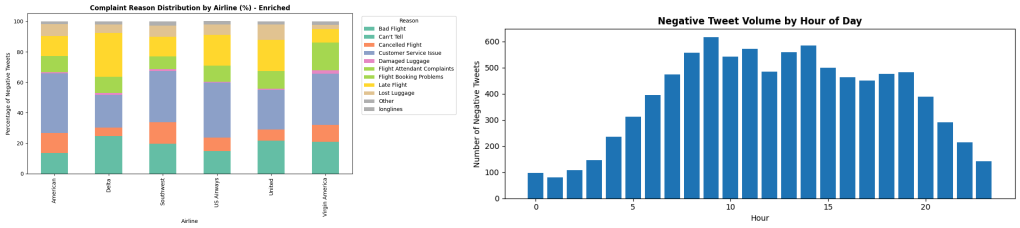

Figure 2. A: Complaint reason by airline distribution (%) (left), B: Negative tweet volume by hour of day

Section 3: Emotion Classification Using a Pretrained Transformer

Using the j-hartmann/emotion-english-distilroberta-base, each tweet was classified into one of seven emotions beyond just “negative sentiment.” Notably, sadness exceeds anger (27% vs. 8.3%), indicating that dissatisfaction is more often classified by disappointment rather than outrage. Joy accounts for 14% of tweets, which signal that early arrivals and friendly service generate appreciation (Figure 3). Customers express opinions more publicly when they are let down rather than when they are satisfied.

Sadness is typically triggered by cancelled flights (68.9%), late flights (38.3%), lost luggage (35.6%), and customer service (33.3%). Anger is typically rooted in customer service (13.8%) and flight attendant complaints (29.4%). Issues in reliability are tied to sadness and unmet expectations whereas breakdown in human interaction looks to trigger anger and frustration becomes personal (Figure 3A).

Across airlines, American and US Airways have the highest sadness (32%) and elevated anger (10%), which matches their higher share of negative sentiment. United has strong sadness (26.6%) and surprise (19.6%) whereas Delta and Southwest are more balanced with lower anger. Virgin has the highest joy (23.4%) and the lowest anger (3.4%).

In terms of hours of the day, sadness and neutral expression are most common during peak travel times (morning through mid-afternoon) whereas anger is relatively stable with slight increases during hours of operation (Figure 3B). Three structural insights I would provide to clients are: 1) customers are expectation-driven when it comes to airlines, and disruption is tied to let downs rather than anger, 2) when

passengers feel ignored, their sadness turns to anger and 3) early arrivals and attentive service brings noticeable joy that can improve the goodwill surrounding an airline brand.

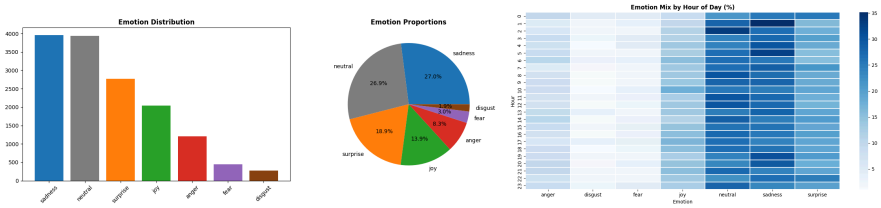

Figure 3. A: Emotion distribution (left), B: Emotions by hours of the day (right)

Section 4: Language & Expression Analysis

It is also important to understand how customers express their emotions. Negative tweets tend to be longer, as sadness has the highest average word count at 19.2 words. Examples illustrate that disappointment and frustration are more detailed ("delayed for hours") whereas joy is more concise and expressive ("great flight!"). Shorter and happier tweets feel more transactional whereas longer tweets must explain unresolved issues. @mention frequency is stable (1.1 mentions per tweet) across airlines. However, anger and sadness usually combine mentions with complain-heavy language ("cancelled") to escalate issues. Emoji usage is highest in joy, fear, and surprise (0.09 per tweet) whereas anger and disgust are lowest, meaning frustration is voiced in words ("rude", "suck") rather than symbols. All caps usage is elevated in surprise, sadness, and anger, signaling heightened emotions in times of urgency. Finally, n-grams analysis confirms that common words associated with emotions are logical (anger and disgust: "worst airline", "rude"; sadness: "flight cancelled", "help"; joy: "love", "great flight"). "Customer service" across all emotions appears to be the dominant bigram, meaning that it is a key component shaping airline experiences of all kinds. For service teams, I would suggest paying attention to long tweets to address concerns, identify signs of anger escalation by key words, and find urgency flags.

Section 5: Recommendations

Across data exploration, the consistent finding is that operational breakdown alone does not drive dissatisfaction, but rather in tandem with poor communication and service. Customer service drives the biggest complaint category (31.7%), and failure typically drives sadness rather than anger. Delays lead to disappointment whereas lack of communication causes anger. The first recommendation is to invest in proactive service measures rather than focusing solely on timely scheduling. Airlines should deploy things such as automated disruption alerts and enable in-app free flight rebooking. When customers feel informed and supported during disruption, sadness is less likely to convert into anger and builds trust with the brand even when inevitable technical issues arise with planes.

The second recommendation is to implement emotion-aware support digesting within social media workflows. Emotion classification and linguistic analysis show that angry tweets are longer, contain stronger negative language ("worst," "rude"), and signal urgency through capitalization. Not all negative tweets require the same response. Airlines should prioritize anger-classified tweets for de-escalation while responding to sadness with empathy. This allows teams to allocate resources more effectively, reduce risk of mass outrage, and create positive brand moments when joy is expressed.

The third recommendation is to strengthen communication during peak operational hours. Negative tweet volume peaks between 8AM and 3PM, aligning with high travel congestion. Rather than attempting to eliminate all delays, airlines should improve transparency through real-time delay updates, visible support wait times, and automated notifications. Clear communication during operational issues reduces uncertainty, prevents emotional escalation, and preserves customer trust.